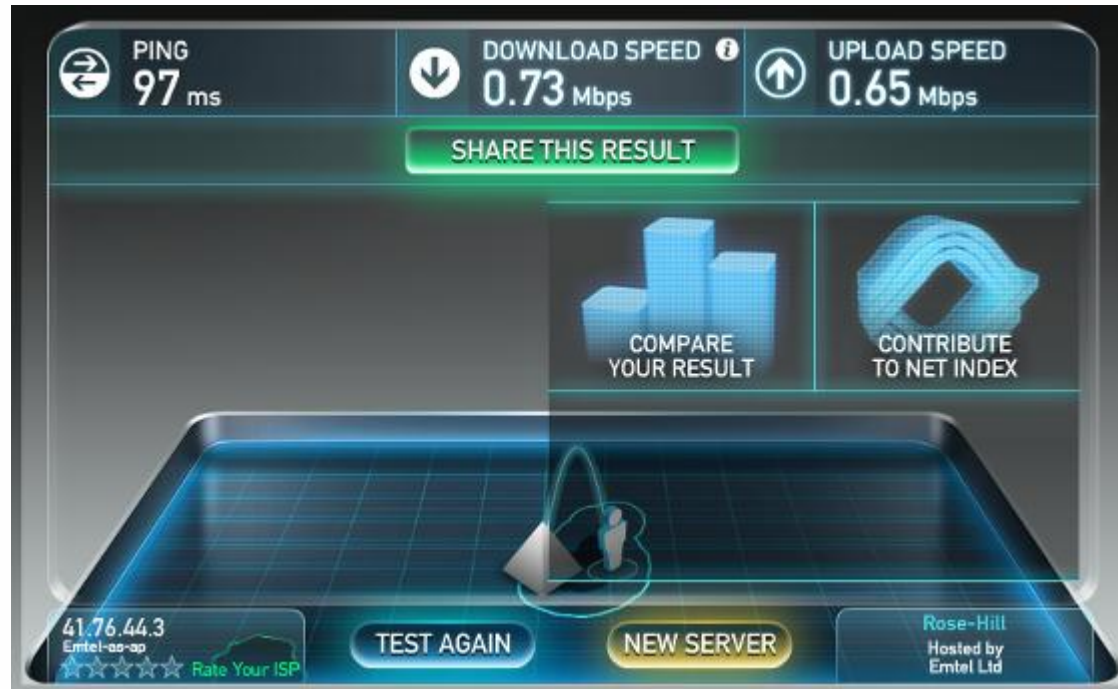




INTERNET SPEED TEST



PRESENTED BY:

B.AKHILESH [24248-CS-027]

M.GANESH [24248-CS-028]

M.ADNAN ALI [24248-CS-029]

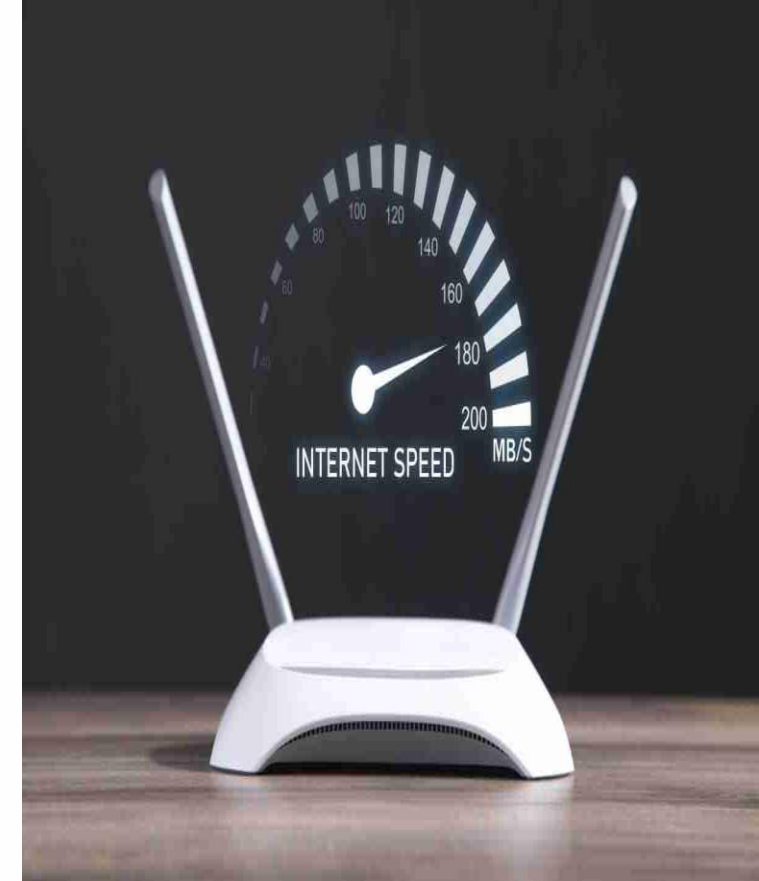
M.GEETALI [24248-CS-031]

S.VISHNU VARDHAN [24248-CS-032]

INTRODUCTION

An **Internet Speed Test** is a simple tool used to check how fast an internet connection is. It measures the **download speed**, **upload speed**, and **ping (latency)** of the network. These values help users know whether their internet connection is working properly and is fast enough for activities like browsing, online classes, video streaming, gaming, and file downloading.

The Internet Speed Test connects to a server, sends and receives data, and calculates the speed of the connection. The results are usually shown in **Mbps (Megabits per second)**. This project helps users monitor their network performance and identify slow internet problems quickly.



OBJECTIVES

- **To measure internet speed**
The main objective is to check the speed of the internet connection by measuring download speed, upload speed, and ping.
- **To monitor network performance**
It helps users know whether their internet connection is working properly or experiencing slow performance.
- **To identify internet issues**
The application helps detect problems such as slow speed, high latency, or unstable connections.
- **To provide accurate results**
It displays internet speed in a clear and easy-to-understand format using Mbps (Megabits per second).
- **To improve user experience**
Users can decide whether their internet is suitable for activities like video streaming, online classes, gaming, video calls, and downloading files.



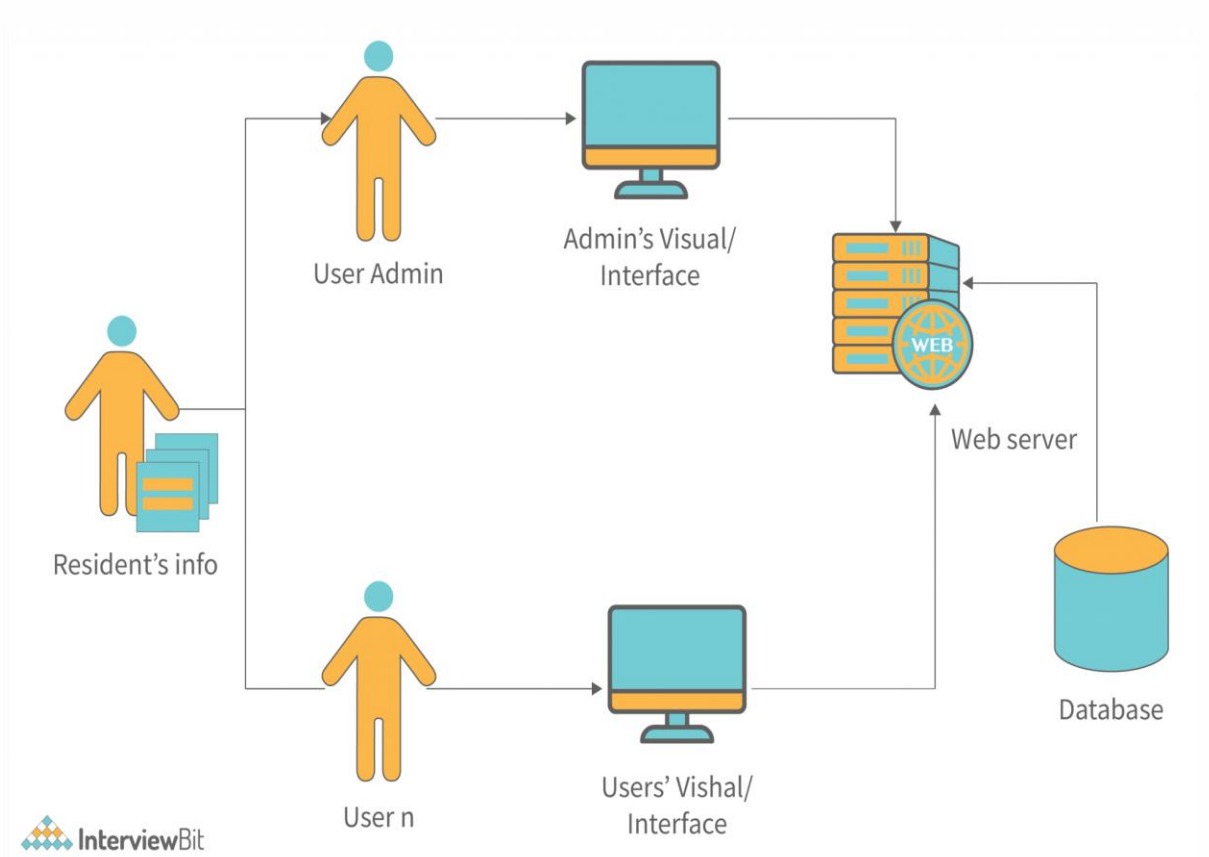
TECNOLOGIES USED

- **Python** – Main programming language used to develop the application.
- **Speedtest-cli** – Measures download speed, upload speed, and ping.
- **Tkinter** – Creates the graphical user interface (GUI).
- **JSON** – Stores speed test results.
- **CSV** – Saves results in Excel-compatible format.
- **Datetime** – Records the date and time of each test.

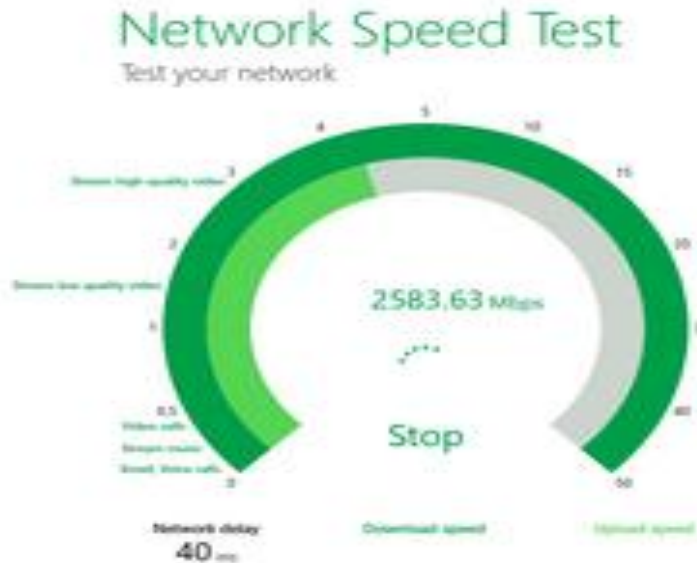


WORKING PROCESS

- **Start the Application** – The user opens the Internet Speed Test application.
- **Connect to Server** – The application connects to the nearest speed test server.
- **Measure Ping** – It checks the response time (latency).
- **Download Test** – It measures the internet download speed.
- **Upload Test** – It measures the internet upload speed
- **. Display Results** – The download speed, upload speed, and ping are shown on the screen.
- **Save Report** – The results are saved in **JSON** or **CSV** format with the current date and time.



OUTPUT SCREEN



FEATURES

- Measures **Download Speed**
- Measures **Upload Speed** Checks
- **Ping (Latency)** Connects to the
- **Nearest Server** Displays
- **Accurate Results Simple and User-Friendly**
- **GUI** Saves Results in **JSON/CSV** Format Records
- **Date and Time** of Each Test
- **Fast and Easy** to Use
- Helps **Monitor Internet Performance**



SYSTEM REQUIREMENTS

Hardware Requirements

- **Processor:** Intel Core i3 or higher
- **RAM:** 4 GB or more
- **Storage:** 100 MB free space
- **Internet Connection:** Active broadband or Wi-Fi

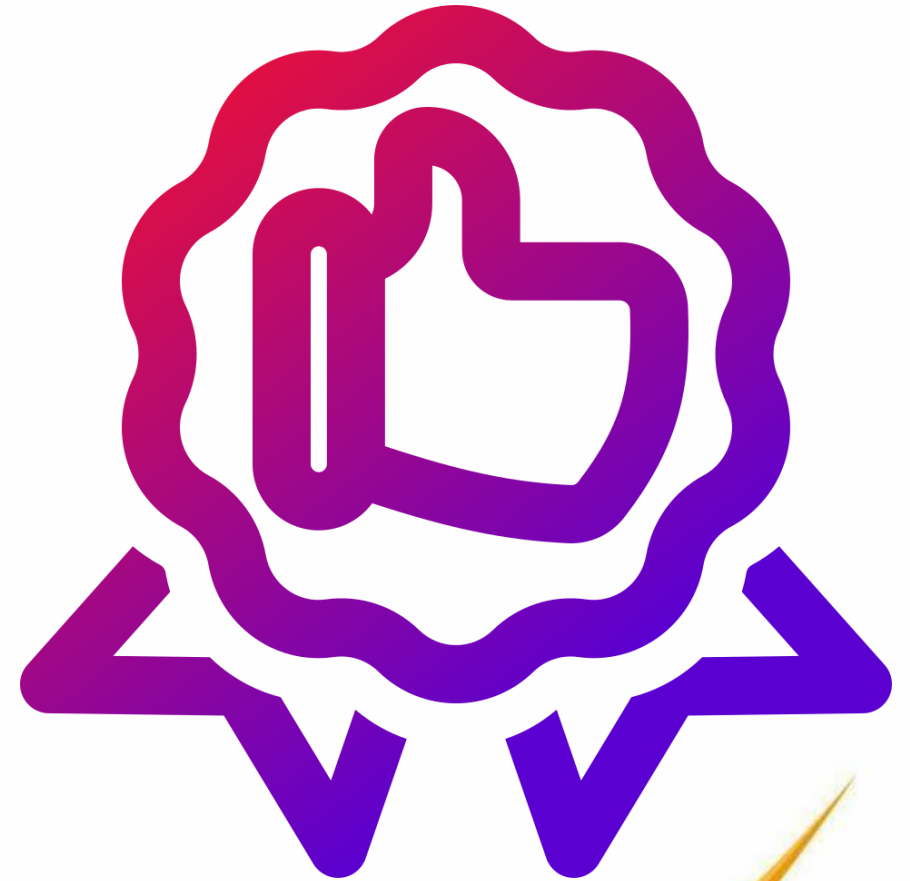
Software Requirements

- **Operating System:** Windows 10/11 (or Linux/macOS)
- **Python:** Version 3.8 or above
- **Libraries:** Speedtest-cli, Tkinter
- **Editor/IDE:** VS Code, PyCharm, or IDLE



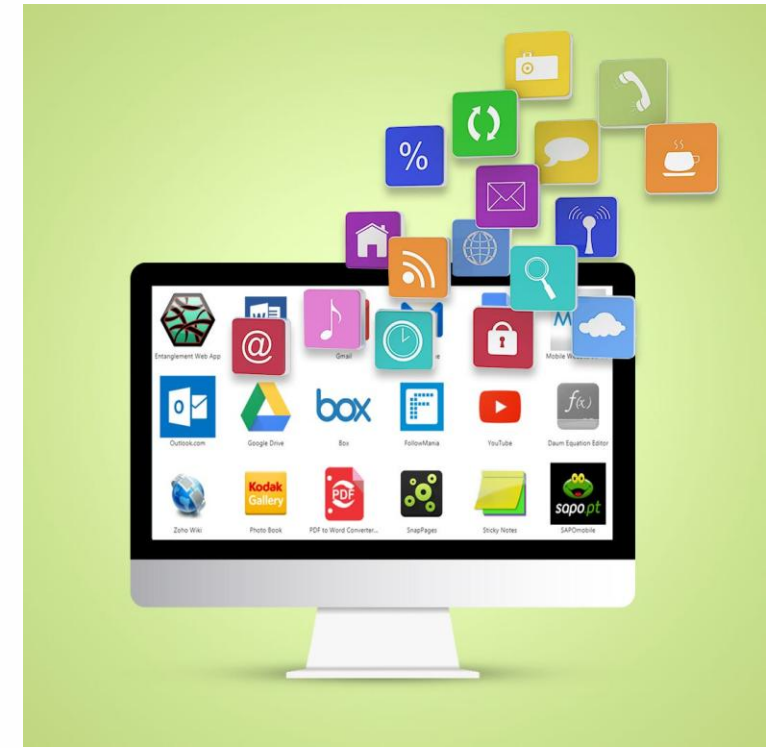
ADVANTAGES

- **Easy to use** with a simple interface.
- **Measures download, upload, and ping** accurately.
- **Helps identify internet speed issue.**
- **Provides quick and reliable results.**
- **Saves test reports** in JSON/CSV format.
- **Monitors network performance** over time.
- **Useful for home, office, and educational purposes.**
- **Supports better troubleshooting** of internet problems.



APPLICATIONS

- **Home Users** – Check internet speed for browsing, streaming, and online classes.
- **Offices** – Monitor network performance for daily work.
- **Educational Institutions** – Test internet quality in schools and colleges.
- **Internet Service Providers (ISPs)** – Verify internet speed for customers.
- **Online Gaming** – Check ping and connection quality.
- **Video Conferencing** – Ensure stable internet for meetings and video calls.
- **Network Troubleshooting** – Identify slow or unstable internet connections.



FUTURE ENHANCEMENTS

- Add **real-time speed monitoring**.
- Display **speed graphs and charts**.
- Support **multiple server selection**.
- Store **test history** in a database.
- Develop a **mobile application**.
- Add **Wi-Fi signal strength** analysis. Generate **detailed PDF reports**.
- Send **alerts** when internet speed drops.



CONCLUSION

The **Internet Speed Test** is a simple and useful application that measures **download speed, upload speed, and ping**. It helps users check the performance of their internet connection and identify network problems quickly. The application is easy to use, provides accurate results, and allows users to save speed test reports for future reference. It is useful for home users, offices, educational institutions, and Internet Service Providers.



THANK YOU